

Python Semester Project

Real-Time Data Plotter

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Real-Time Data

From Coinbase API for Bitcoin

Get product ticker - snapshot
information about the last trade

Get product stats - statistics (last
24 hours)

- Included lines for price high and price low to the live plot



Animated Line Plot

Constructed and animated figure using the matplotlib package.

Animate function handles line updates for BTC price. Constructs constant lines for high and low price comparison.

```
def animate(frame,xs,ys,h_list,l_list):  
    data = get_data()  
    xs.append(data.time)  
  
    #limit length of time (x) list  
    xs=xs[-x_len:]  
    ax.set_xlim(xs[0],xs[-1])  
  
    #limit length of rate (y) lists  
    ys.append(data.price)  
    ys=ys[-x_len:]  
  
    #adjust y limits for new data point  
    y_limits = ax.get_ylim()  
    up_lim = y_limits[1]  
    low_lim = y_limits[0]  
  
    if(data.price < low_lim):  
        low_lim = data.price-100  
    if(data.price > up_lim):  
        up_lim = data.price+100  
  
    ax.set_ylim(low_lim, up_lim)  
  
    ax.set_xticks(xs)  
    plt.xticks(rotation=45, ha='right')  
  
    h_list.append(high)  
    l_list.append(low)  
  
    h_list = h_list[-x_len:]  
    l_list = l_list[-x_len:]  
  
    #update line  
    line.set_data(xs,ys)  
    h_line.set_data(xs,h_list)  
    l_line.set_data(xs,l_list)
```